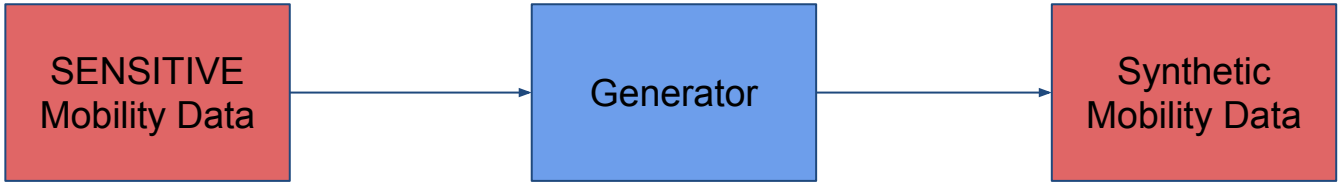
Synthetic Mobility Data:
A User-Level Membership Inference Assessment

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Motivation

The Issue: Lack of open access mobility dataset due to privacy concerns



- Home location
- Work location
- Regular travel schedule

Details obscured

The idea: generate & publish a version of the dataset with privacy protected

Research Questions

RQ1: Can generative models accurately reproduce origin–destination mobility patterns from real trip data?

Utility

RQ2: To what extent do these models leak information about individual users through membership inference attacks?

Privacy

RQ3: Are users with more trips at a greater risk of privacy leakage than less active users?

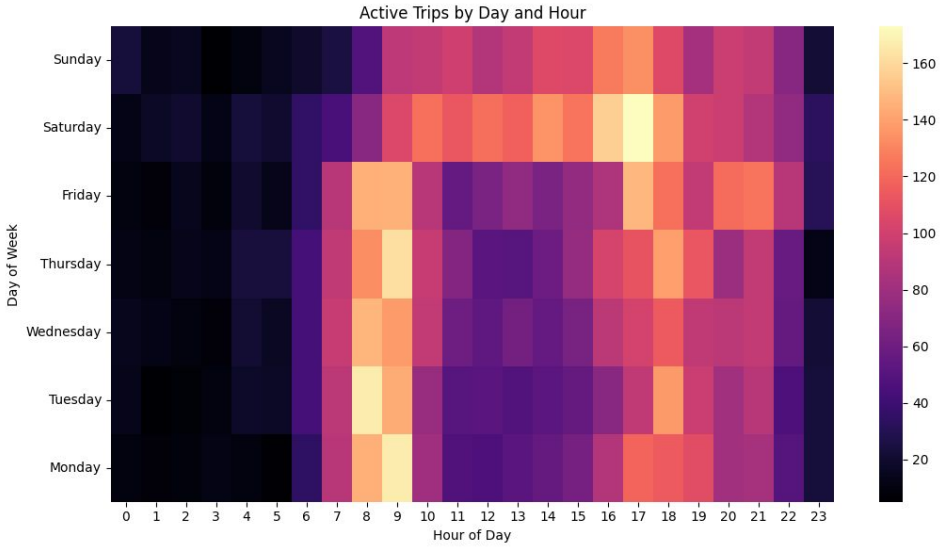
Privacy by Activity Group

Dataset Introduction

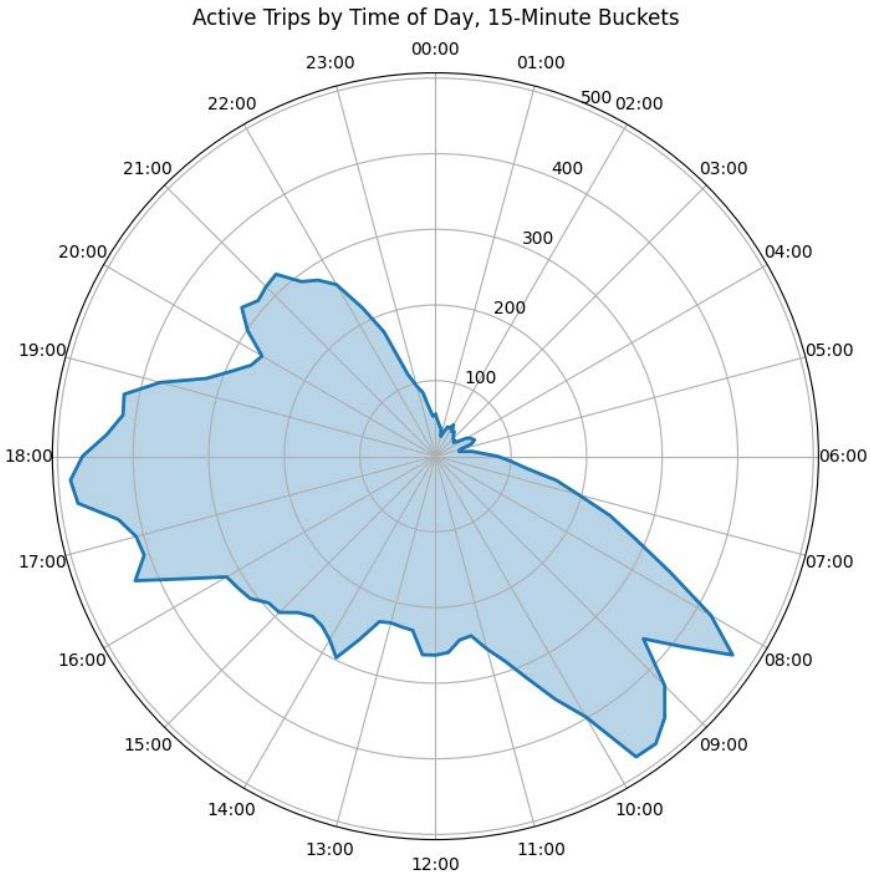
GeoLife Dataset (Microsoft Asia Research)

GPS Trajectories → OD trips segmentation

7,739 trips across 154 users
(average trip being 28.7 minutes & 13.6km)

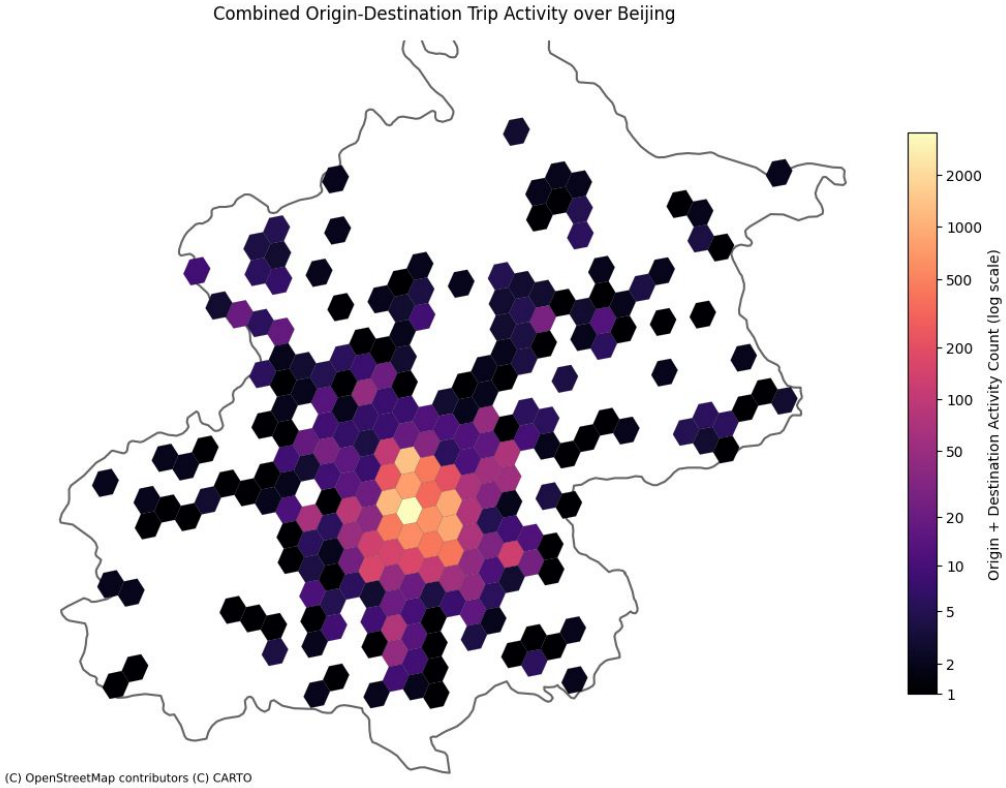


Clear schedule change on weekends

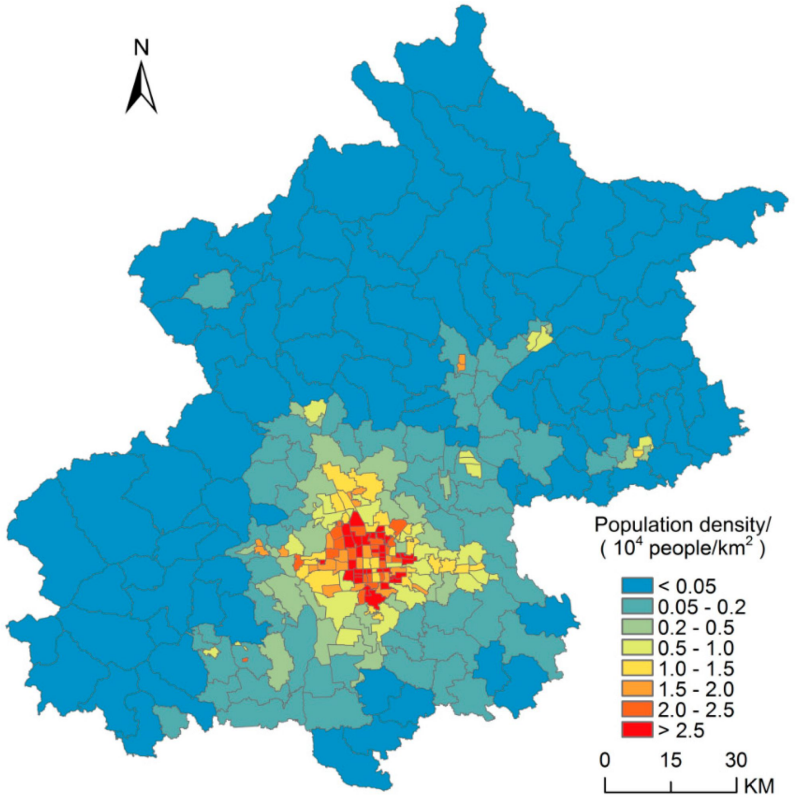


Clear ‘rush hour’ peaks

Dataset Introduction



Dataset OD Coverage

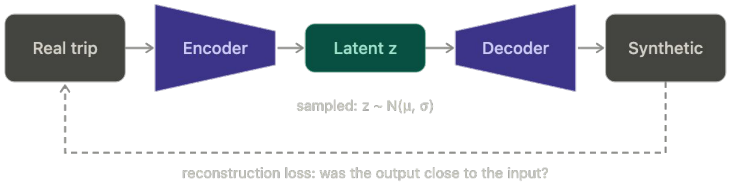


Population Density [2]

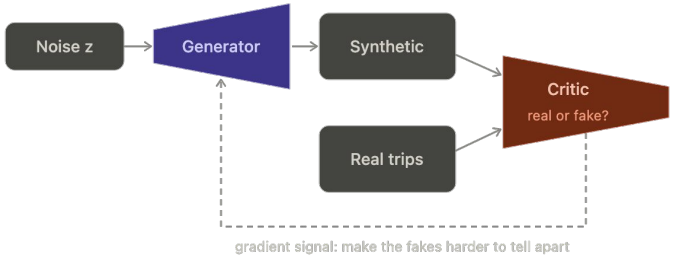
[2] Population Spatialization in Beijing City Based on Machine Learning and Multisource Remote Sensing Data - Miao He,Yongming Xu, Ning Li, 2020

Models

1) **VAE** (Variational Autoencoder) — reconstruction loss + KL loss



2) **WGAN-GP** (Wasserstein GAN with Gradient Penalty) — generator vs. critic adversarial game



Benchmark) **Bootstrap** — resampling real trips directly with replacement



Operating on transformed features (continuous OD coordinates, trip distance/duration, hour/day-of-week)
Manual hyperparameter search

Utility Evaluation

Similarity Comparison

Compares discrete marginals (Jensen–Shannon distance) between real and synthetic dataset. Locational attributes are banded into grid cells and temporal attributes are put into hour and day of week,
0 = identical distributions, 1 = no overlap at all.

Downstream Task

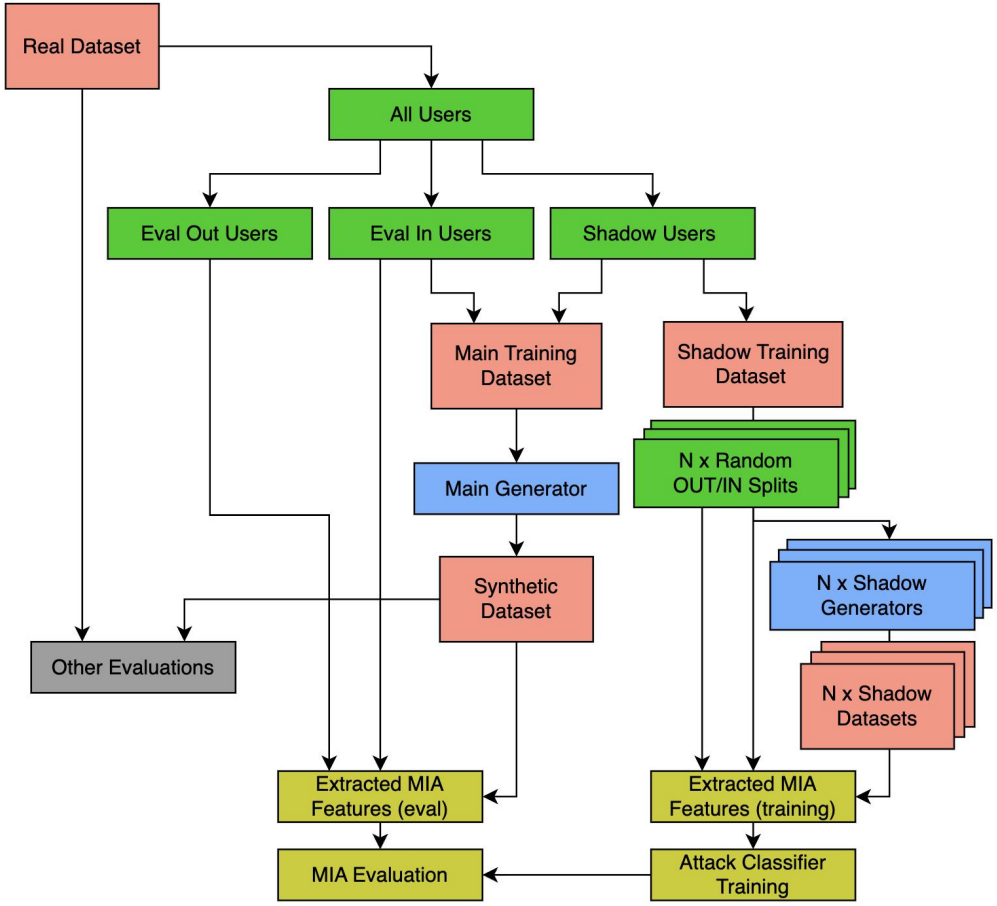
Train a model (random forest) to predict a trip's destination cell from its origin coordinate, using duration and distance as well. Trained on Synthetic dataset and tested on real data set, this is then normalised by performance score when using the holdout subset of real data.

1.0 = synthetic is a perfect substitute; 0 = useless.

Membership Inference

Was this real user used to generate the synthetic dataset or not?

High accuracy to the (Yes/No) inference question → Privacy Leak



Membership Inference

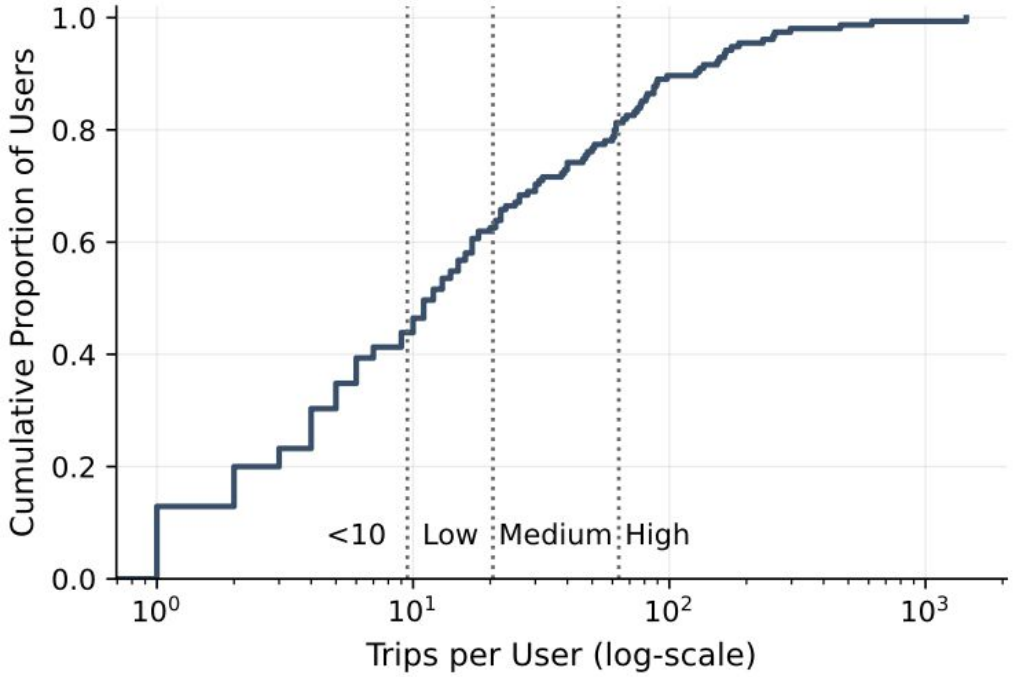
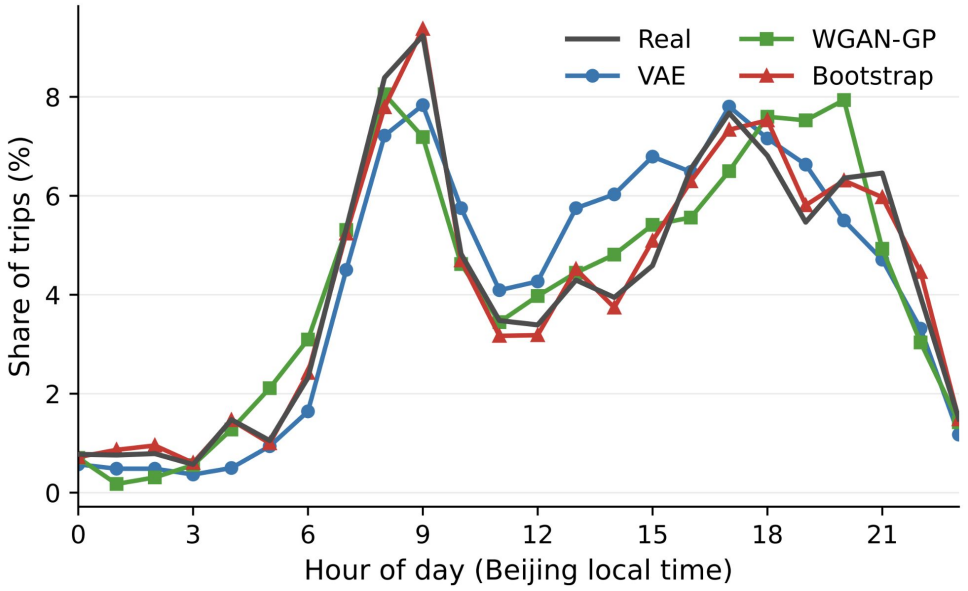
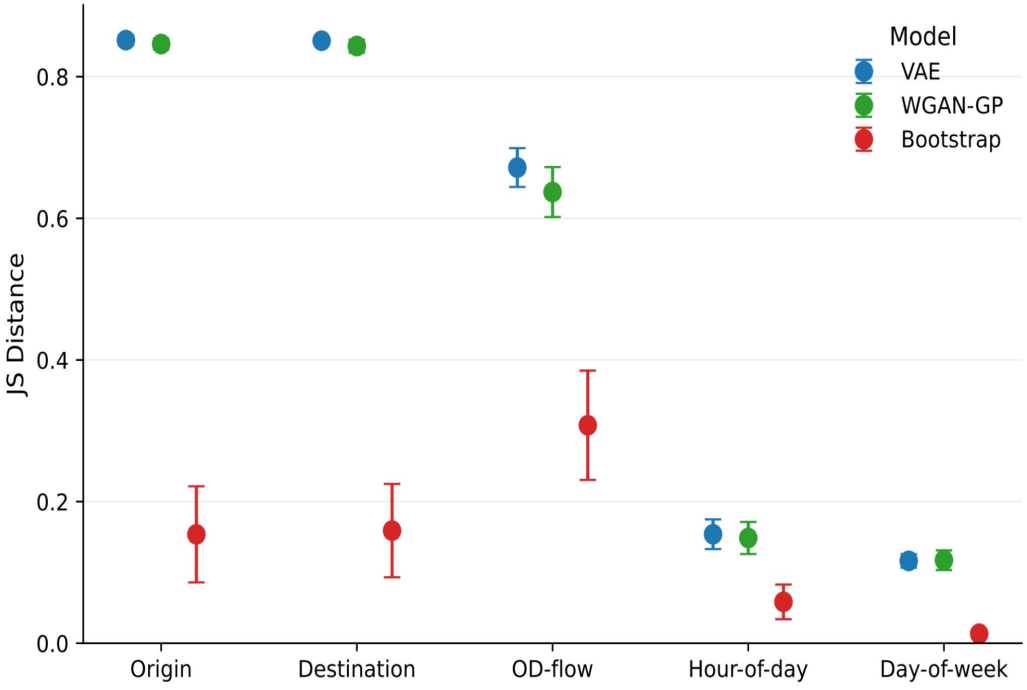


Table 6: User activity group, defined by total trip count per user.

Group	Trips/user	Description	Users
MIA Excluded	≤ 9	Too few trips for reliable membership inference (non-eligible)	67
Low	10–20	Lower third of eligible users	29
Medium	21–63	Middle third of eligible users	29
High	≥ 64	Upper third of most active eligible users	29
Total			**154**

RQ1 - Utility





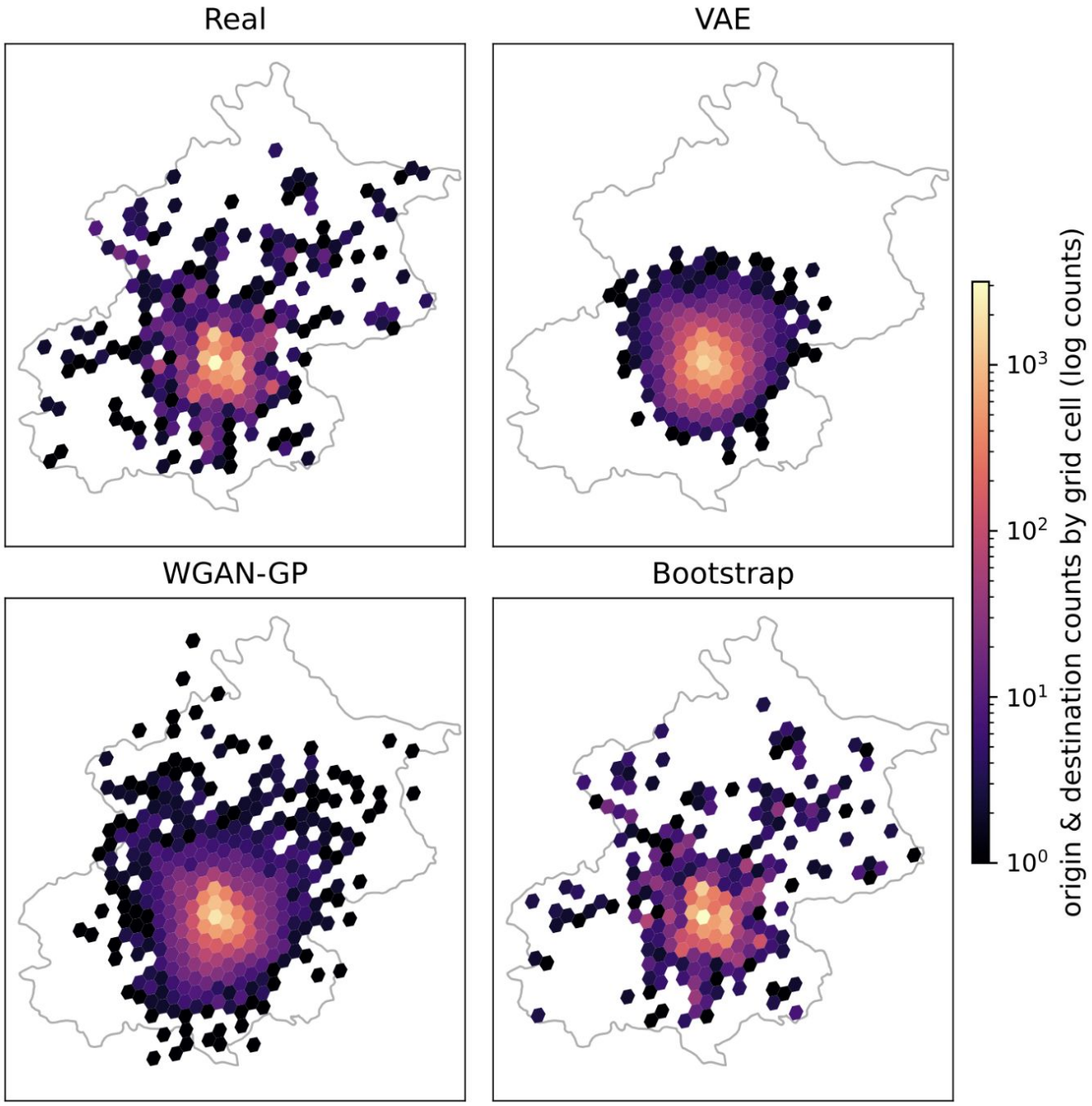
Temporal structure reproduced well (realistic trip start times)

JS distance 0.12–0.15 for hour-of-day/day-of-week

Results

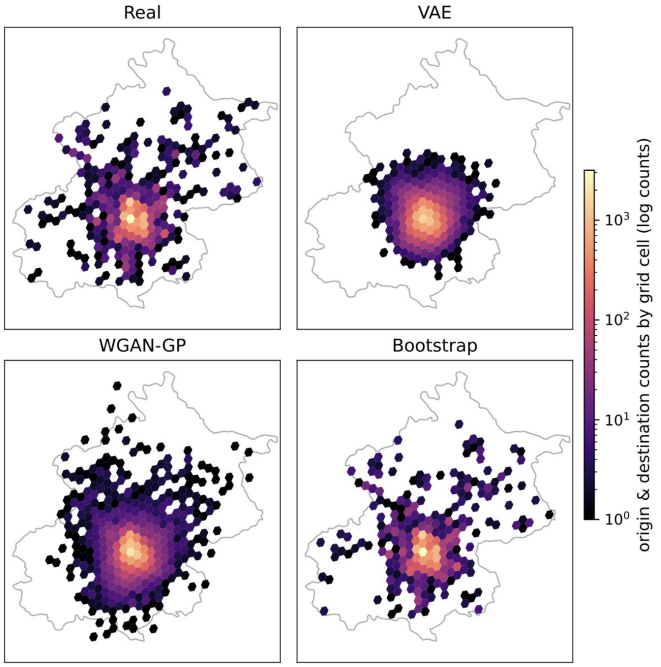
Overgeneralization to unobserved locations (even outside of Beijing)

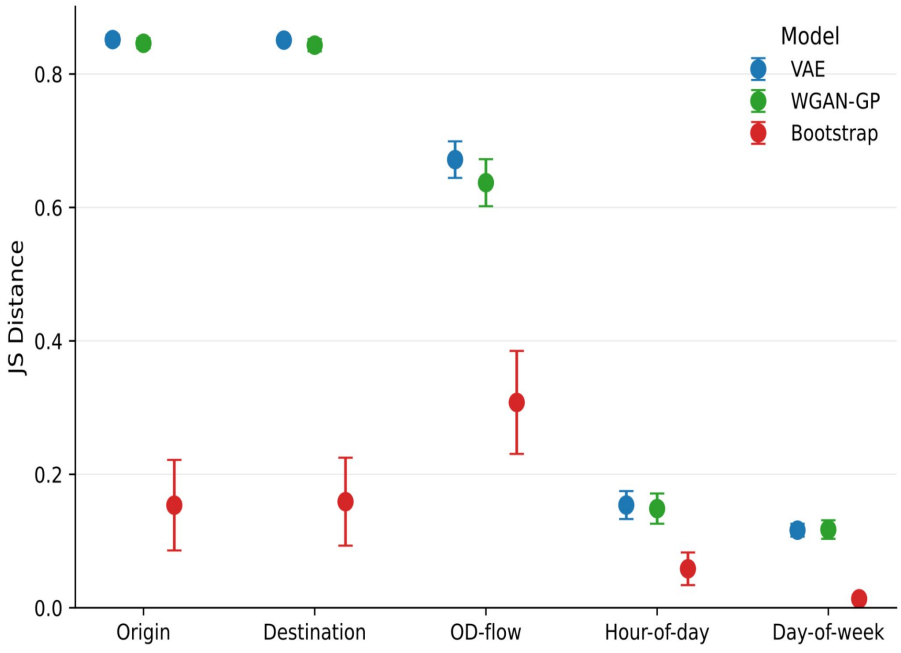
WGAN-GP preserved real spatial dispersion better; VAE compressed trips toward the urban centre

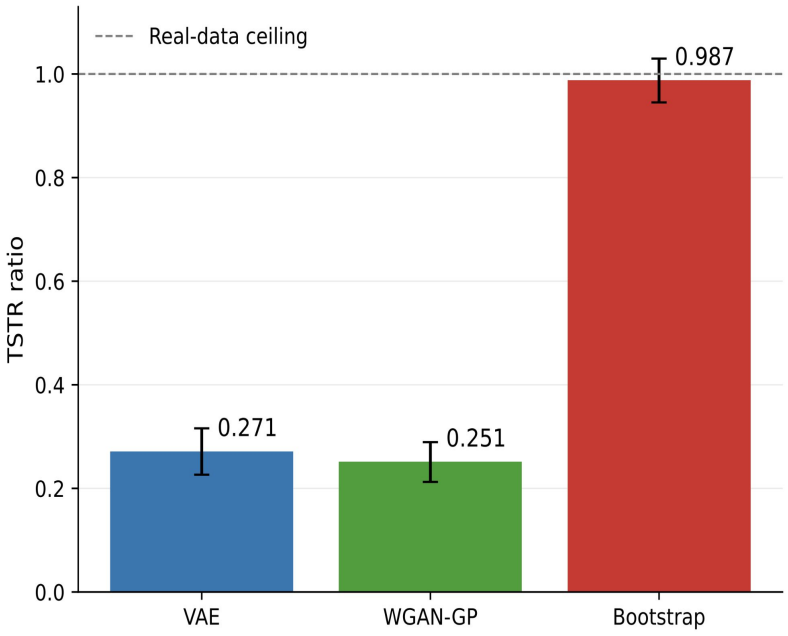


RQ1 - Utility

Spatial structure reproduced poorly
(JS distance >0.8; TSTR ratio 0.25–0.27 vs. 0.99 for the bootstrap)

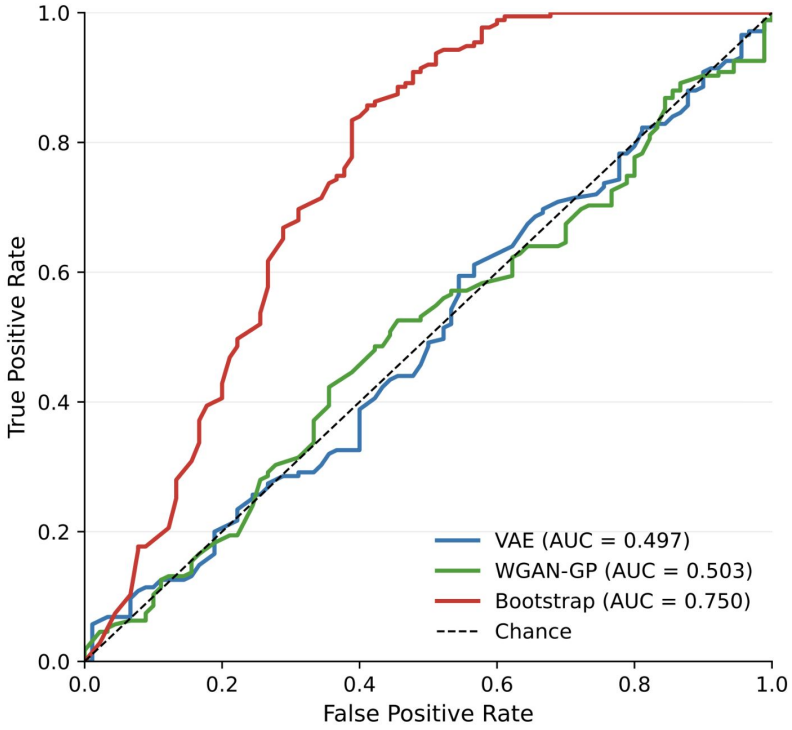


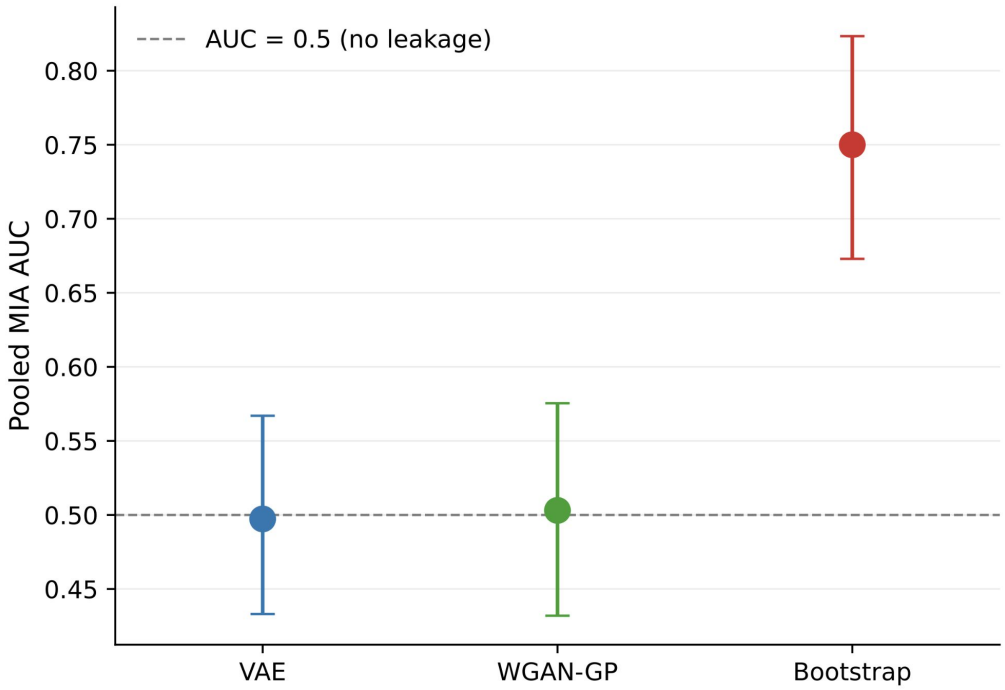




RQ2 Results - MIA

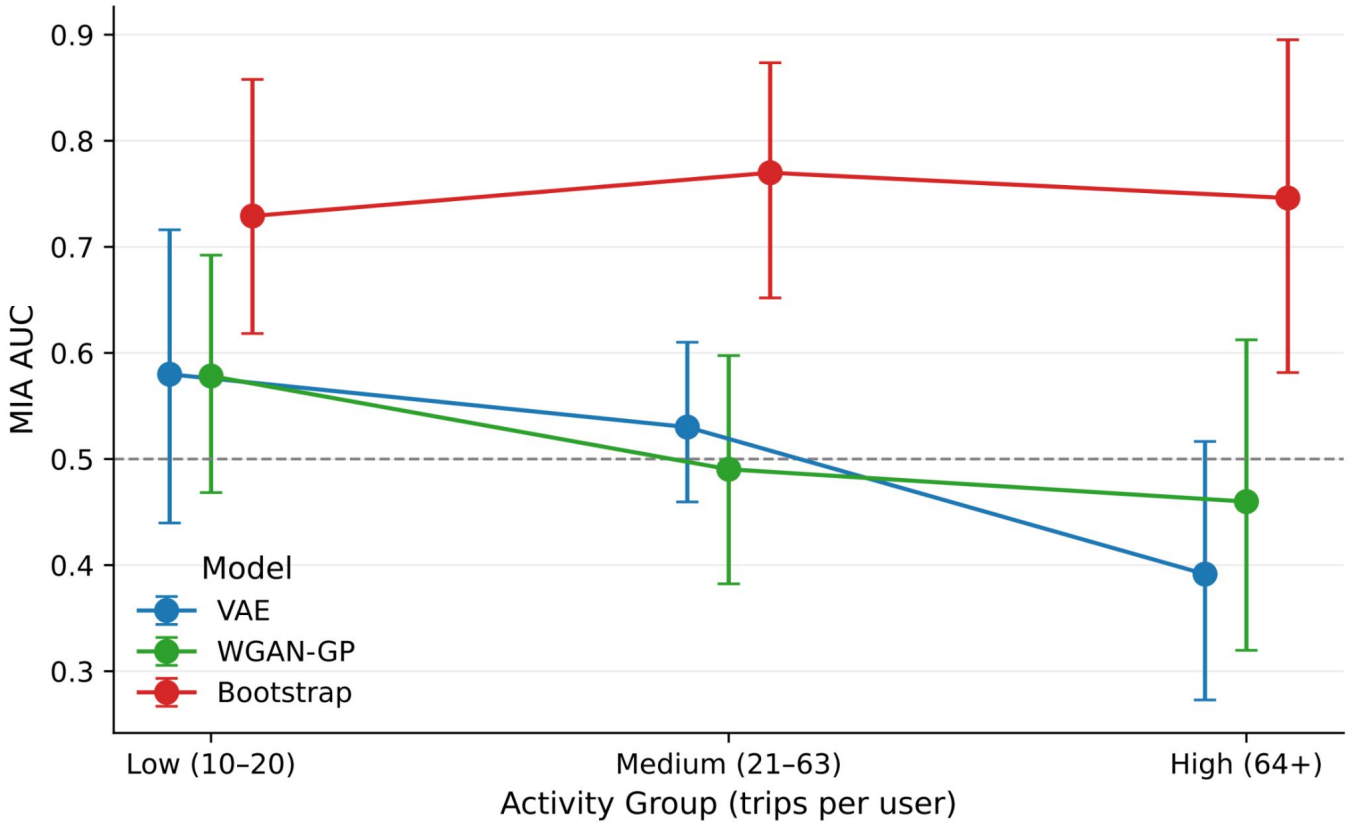
Bootstrap positive control confirmed a working attack (AUC 0.75)





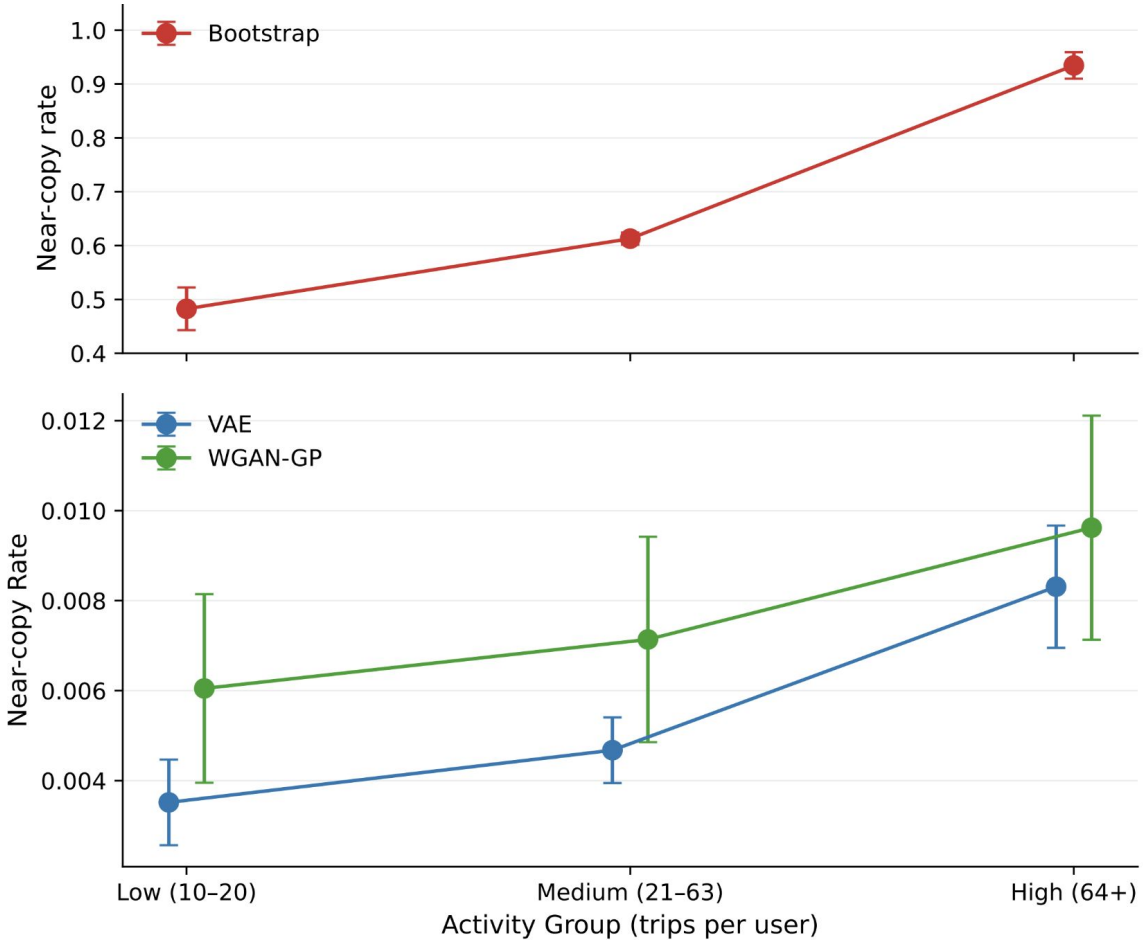
Neither generative model showed measurable leakage (AUC 0.50)

RQ3 Results - By Activity Group



MIA-by-activity-group was inconclusive:
as few as 28 users per group gave confidence intervals too wide to support any claim

RQ3 Results - By Activity Group



Near-copy rate rose monotonically with activity across all models (including the bootstrap) indicating higher-activity users carry greater record-reproduction risk in principle

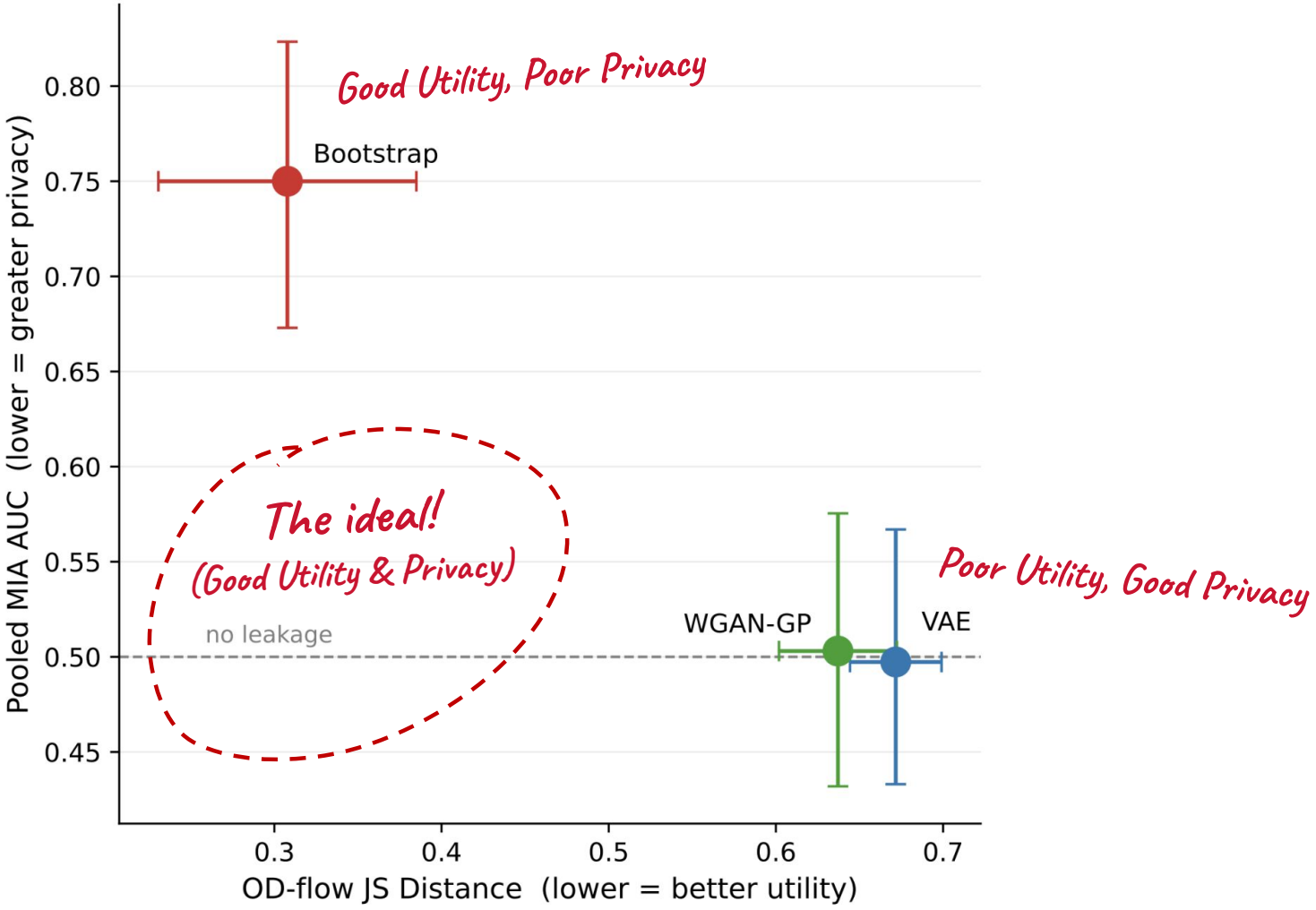
Further Research

1. **Learned embeddings** as an alternative spatial representation (vs. raw coordinates), which is better suited to the discrete nature of real-world travel behaviour, which should addresses the over-generalisation to unobserved locations.

2. **Use a larger dataset** to give greater statistical power (particularly for RQ3: activity-stratified MIA).

3. **Additional models,** both architectures and training variants to trace the trade-off region between the bootstrap and deep-model unfitting extremes, rather than only the two endpoints.

Further Research



Conclusions

Central finding: strong privacy achieved at substantial cost to utility (no observed decoupling)

RQ1 (Utility):

- Temporal structure reproduced well (JS distance 0.12–0.15 for hour-of-day/day-of-week; realistic trip start times and durations)
  Spatial structure reproduced poorly (JS distance >0.8; TSTR ratio 0.25–0.27 vs. 0.99 for the bootstrap)
- WGAN-GP preserved real spatial dispersion better; VAE compressed trips toward the urban centre
- Both models generalised to unobserved locations (over-generalisation)

RQ2 — Privacy

- Bootstrap positive control confirmed a working attack (AUC 0.75); its sub-1.0 ceiling reflects ambiguity from OD discretisation shared across individuals
- Neither generative model showed measurable leakage (AUC ≈0.50)

RQ3 — Activity-level risk:

- MIA-by-activity-group was inconclusive: as few as 28 users per group gave confidence intervals too wide to support any claim
- Near-copy rate rose monotonically with activity across all models, including the bootstrap, indicating higher-activity users carry greater record-reproduction risk in principle


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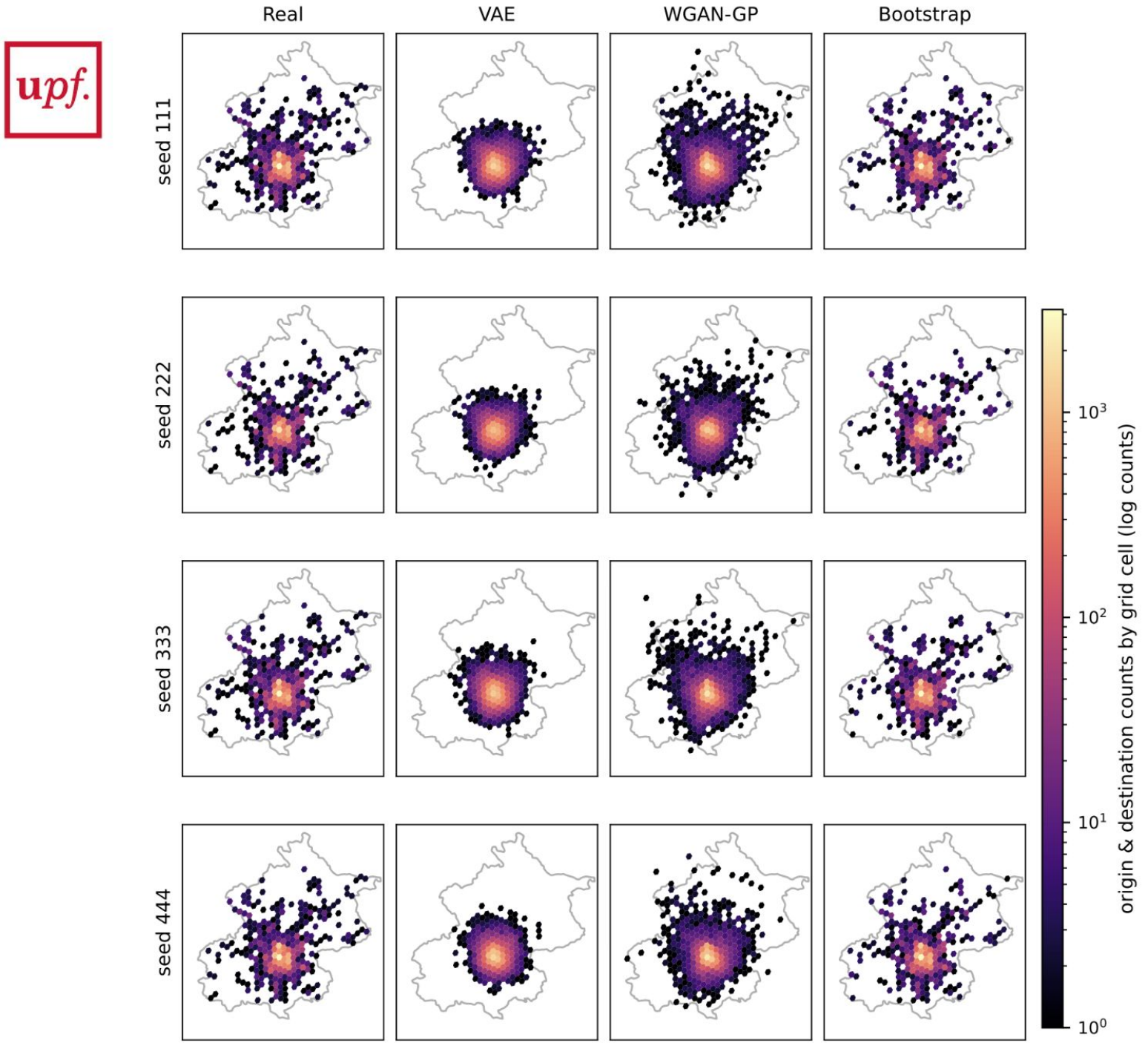
References

Shadow-model attack paradigm — the method for RQ2: train shadow models on known IN/OUT splits, learn a membership classifier from their outputs. → Shokri, R. et al. (2017). *Membership Inference Attacks against Machine Learning Models*. arXiv:1610.05820.

MIA adapted to generative models — closest prior work; attacker sees only a synthetic dataset, reduced to a fixed-length feature vector. → Stadler, T., Oprisanu, B. & Troncoso, C. (2022). *Synthetic Data – Anonymisation Groundhog Day*. arXiv:2011.07018.

The VAE → Kingma, D.P. & Welling. *Auto-Encoding Variational Bayes*. arXiv:1312.6114.

WGAN-GP → Gulrajani, I. et al. (2017). *Improved Training of Wasserstein GANs*. arXiv:1704.00028


Real
VAE
WGAN-GP
Bootstrap
seed 111
seed 222
seed 333
seed 444
origin & destination counts by grid cell (log counts)
10^0
10^1
10^2
10^3

Model Parameters

Table 2: The ten-dimensional continuous feature vector supplied to each generative model.

Feature	Description	Transformation
Origin latitude	Start centroid latitude	Standardised
Origin longitude	Start centroid longitude	Standardised
Destination latitude	End centroid latitude	Standardised
Destination longitude	End centroid longitude	Standardised
Duration	Trip duration (minutes)	Log, then z-scored
Distance	Haversine OD distance	Log, then z-scored
Hour (sine)	Start hour of day, sin term	Cyclical, 24h
Hour (cosine)	Start hour of day, cos term	Cyclical, 24h
Day-of-week (sine)	Start day of week, sin term	Cyclical, 7d
Day-of-week (cosine)	Start day of week, cos term	Cyclical, 7d

Model Parameters

Table 3: Final hyperparameters for the generative models.

Setting	**Description**	**Value**
Shared		
Optimiser	Adam	—
Batch size	Samples per batch	64
VAE		
Latent dimension	Size of latent space	8
Learning rate	Adam step size	1×10^{-3}
β	Final KL weight	0.1
KL annealing	Warm-up fraction of epochs	0.4
Epochs	Training epochs	80
WGAN-GP		
Noise dimension	Generator input dimension	64
Hidden dimension	Base width of generator/critic	128
Learning rate	Adam step size	1×10^{-4}
λ_{gp}	Gradient penalty weight	10
Critic steps	Critic updates per generator update	3
Adam (β_1, β_2)	Adam momentum terms	$(0.5,\ 0.9)$
Epochs	Training epochs	100

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	VAE	WGAN-GP	Bootstrap
Origin JS Distance	0.852 ± 0.007	0.846 ± 0.008	0.154 ± 0.068
Destination JS Distance	0.851 ± 0.003	0.843 ± 0.009	0.159 ± 0.066
OD-flow JS Distance	0.672 ± 0.027	0.637 ± 0.035	0.308 ± 0.077
OD-flow Coverage	0.427 ± 0.018	0.372 ± 0.013	0.673 ± 0.050
Hour-of-day JS Distance	0.154 ± 0.021	0.149 ± 0.023	0.058 ± 0.024
Day-of-week JS Distance	0.116 ± 0.010	0.117 ± 0.014	0.013 ± 0.006
Correlation Difference	0.071 ± 0.013	0.052 ± 0.019	0.030 ± 0.010
Wasserstein Distance	0.167 ± 0.002	0.120 ± 0.005	0.018 ± 0.004
Spread Ratio	0.804 ± 0.011	0.908 ± 0.024	1.007 ± 0.012
Kolmogorov-Smirnov Distance	0.166 ± 0.012	0.157 ± 0.014	0.219 ± 0.017
Forecast Task (TSTR)	0.271 ± 0.045	0.251 ± 0.038	0.987 ± 0.042

Table 7: RQ1 utility results (mean $\pm$ standard deviation over five seed runs). JS Distance denotes the Jensen-Shannon distance between the corresponding real and synthetic marginal distributions. Mean Wasserstein Distance is computed on features standardised using statistics from the real data

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	VAE	WGAN-GP	Bootstrap
Users Evaluated	85	85	85
Balanced Accuracy	0.515 [0.460, 0.571]	0.474 [0.417, 0.529]	0.692 [0.629, 0.757]
AUC	0.497 [0.433, 0.567]	0.503 [0.432, 0.575]	0.750 [0.673, 0.823]
Maximum Advantage	0.050 [0.032, 0.177]	0.070 [0.020, 0.206]	0.446 [0.350, 0.577]
TPR @10% FPR	0.114 [0.062, 0.194]	0.103 [0.038, 0.184]	0.187 [0.078, 0.386]
TPR @5% FPR	0.069 [0.039, 0.135]	0.059 [0.023, 0.116]	0.081 [0.018, 0.237]
TPR @1% FPR	0.000 [0.000, 0.087]	0.030 [0.010, 0.070]	0.013 [0.002, 0.074]
Near-copy Rate	0.009 ± 0.001	0.010 ± 0.002	1.000 ± 0.000

Table 8: RQ2 privacy results. MIA-associated metrics are pooled across seed runs and reported with 95% cluster-bootstrap confidence intervals. Users Evaluated denotes the number of distinct users contributing to each pooled estimate. Near-copy Rate is reported as mean $\pm$ standard deviation over seed runs.

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Model	Activity Group	Users	AUC	TPR @ 10% FPR	Near-copy Rate
VAE	Low (10–20)	28	0.580 [0.440, 0.716]	0.186 [0.057, 0.379]	0.004 ± 0.001
VAE	Medium (21–63)	29	0.530 [0.460, 0.610]	0.263 [0.113, 0.362]	0.005 ± 0.001
VAE	High (64+)	28	0.392 [0.273, 0.517]	0.102 [0.000, 0.193]	0.008 ± 0.001
WGAN-GP	Low (10–20)	28	0.578 [0.468, 0.692]	0.220 [0.024, 0.400]	0.006 ± 0.002
WGAN-GP	Medium (21–63)	29	0.490 [0.382, 0.598]	0.193 [0.029, 0.327]	0.007 ± 0.002
WGAN-GP	High (64+)	28	0.460 [0.320, 0.612]	0.068 [0.016, 0.196]	0.010 ± 0.002
Bootstrap	Low (10–20)	28	0.729 [0.618, 0.858]	0.237 [0.019, 0.586]	0.483 ± 0.040
Bootstrap	Medium (21–63)	29	0.770 [0.652, 0.874]	0.316 [0.036, 0.569]	0.613 ± 0.011
Bootstrap	High (64+)	28	0.746 [0.582, 0.895]	0.237 [0.038, 0.623]	0.935 ± 0.025

Table 9: RQ3 privacy by activity group. MIA-associated metrics are pooled within each activity group and reported with 95% cluster-bootstrap confidence intervals. Users denotes the number of distinct users contributing to each pooled estimate. Near-copy Rate is reported as mean ± standard deviation over seed runs.